Name of the Student: _____ **Roll no:** _____

CLASS: - T.E (CSE) **Division:** **Course: - PCC305CSE- Machine Learning Lab**

Experiment No. 11

**** Reinforcement Learning: Q-Learning, Exploration–Exploitation and Optimal Policy****

Marks:

Date of Performance: / /2026 **Sign with Date:**

Aim: To evaluate the performance of Reinforcement Learning using the Q-Learning algorithm on a simple Grid World decision-making problem, with emphasis on exploration–exploitation strategies and the optimal policy.

Lab Objective: Enable students to understand Q-Learning, implement an agent in a simple Grid World, compare exploration–exploitation strategies such as ε-greedy, evaluate learning performance using cumulative reward and success rate, and interpret the learned optimal policy.

Environment: Simple 4 × 4 Grid World. The agent starts at a fixed start state, moves Up/Down/Left/Right, receives a positive reward on reaching the goal, and receives a small negative step reward. Invalid moves keep the agent in the same state.

Why Grid World? It is compact, easy to visualize, and clearly demonstrates how an agent learns state–action values through trial and error. It also makes the learned optimal policy easy to interpret.

Tools / Libraries Required

- Python 3.x
- Jupyter Notebook / Google Colab
- NumPy
- Matplotlib
- Pandas (optional for result tables)

Theory

Reinforcement Learning (RL) is a machine learning approach in which an agent learns to make decisions by interacting with an environment. At each step, the agent observes a state, selects an action, receives a reward, and moves to a new state. The objective is to learn a policy that maximizes the expected cumulative reward.

1. Q-Learning

Q-Learning is a model-free, off-policy reinforcement learning algorithm. It learns a Q-value Q(s,a),

representing the expected long-term reward for taking action a in state s and then following an effective policy. The Q-table is updated using the Bellman equation:

$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$

where α is the learning rate, γ is the discount factor, r is the immediate reward, s' is the next state, and a' represents a possible next action.

2. Exploration–Exploitation

The agent faces a trade-off between exploration and exploitation. Exploration means trying different actions to discover potentially better rewards. Exploitation means selecting the action currently believed to be best. An ε-greedy strategy balances the two: with probability ε the agent chooses a random action; otherwise it chooses the action with the highest Q-value.

3. Optimal Policy

A policy defines the action selected in each state. After training, the greedy policy is obtained by selecting the action with the maximum learned Q-value in every state. In the Grid World, the optimal policy should guide the agent along a shortest or highest-return path from the start state to the goal while avoiding unnecessary moves.

4. Performance Evaluation

Performance can be evaluated using cumulative reward per episode, success rate, average steps to reach the goal, and convergence of the learned Q-values. A successful agent should obtain higher rewards, reach the goal more consistently, and converge toward a stable policy.

Algorithm

Input: Grid World environment, learning rate α, discount factor γ, exploration rate ε, number of training episodes.

Output: Learned Q-table, cumulative rewards, success rate, and optimal policy.

1. Start.
2. Define the Grid World states, available actions, start state, goal state, and rewards.
3. Initialize the Q-table Q(s,a) with zeros for all state–action pairs.
4. Set α, γ, ε, and the number of training episodes.
5. 5. For each episode, reset the agent to the start state.
6. Select an action using the ε-greedy exploration–exploitation strategy.
7. Execute the action, observe the reward and next state.
8. Update Q(s,a) using the Q-Learning update equation.
9. Set the current state to the next state and continue until the goal is reached or the episode limit is exceeded.

10. Record the cumulative reward, number of steps, and whether the goal was reached.

11. Repeat the training process for all episodes.

12. Derive the optimal policy by selecting the action with maximum Q-value for each state.

13. Plot learning performance such as episode reward and success rate.

14. 14. Compare the effect of different ε values on exploration, convergence, and final performance.

15. Stop.

Expected Observation

At the beginning of training, the agent explores many actions and the cumulative reward may be low or variable. As training progresses, Q-values are updated and the agent increasingly exploits actions with higher estimated returns. The success rate should generally improve, and the learned policy should converge toward actions that lead efficiently from the start state to the goal.
The exact numerical results depend on the random seed, learning parameters, number of episodes, and implementation. Students should report the values produced by their execution rather than assuming fixed values.

Sample Results / Observation

Strategy / ε	Avg. Reward	Success Rate (%)	Avg. Steps	Observation
High exploration (ε=0.5)	_____	_____	_____	More exploration; slower convergence may occur.
Balanced (ε=0.1)	_____	_____	_____	Usually provides a good exploration–exploitation balance.
Low exploration (ε=0.01)	_____	_____	_____	More exploitation; may converge quickly but can miss better actions.

Result

The Q-Learning agent was trained on the Grid World environment using an ε-greedy exploration–exploitation strategy. The final Q-table and greedy policy were obtained after training. The best strategy based on the selected performance criteria was _____. The learned policy reached the goal with a success rate of _____% and an average reward of _____.

Questions:

1. What is Reinforcement Learning, and how is it different from supervised learning?

2. Explain Q-Learning and the meaning of Q(s,a).

3. Write and explain the Q-Learning update equation.

4. What is the exploration–exploitation trade-off?

5. Explain how the ε-greedy strategy works.

Attendance and Timely Submission (10)	**Execution (10)**	**Observations and Records (10)**	**Journal Documentation (10)**	**Oral (10)**